

PROJECT.revive

. an interactive thesis website by mehul sathya narayan

Where linear planning meets organic reality through interactive dual-mode storytelling



. rethinking informality in informal settlements



.introduction

The term “revive” means to return to consciousness or life. This project seeks to intervene in an area in the city of Delhi that has long been ignored, and has gradually declined further to a point where the existing urban fabric of the city has been heavily affected due to problems that exist within this region. My aim is to intervene in this region in two phases, first being the urban re-design and planning of this area at a macro scale, and then focusing on re-developing a certain core area of this region that can be used as a model for replication throughout this region as well as other similar regions spread throughout the city.

.slums : the definition

The word “Slum” refers to informal settlements within cities that have inadequate housing and squalid, miserable living conditions.

Further diving into the definition of a slum:

- 1.6 Billion people live without adequate shelter
- 1 in 7 people on the planet live in a slum
- 1 in 3 urban residents live in slums in developing countries
- In some countries, 90% of the urban population lives in slums.

Major developing cities where slums are one of the main types of housing are Kibera in Nairobi, Manila, and New Delhi. Major characteristics of slums include:

- Unsafe & unhealthy homes (lack of windows, dirt flooring, leaky walls and roofs)
- No access to basic services: water, toilets, transportation.
- Unstable homes and no secure land tenure.



Kibera, Nairobi, KENYA

60% of Nairobi's population lives in slums and around 250,000 people in Kibera making it the biggest slums in Africa and one of the biggest in the world. Most of the structures are made out of mud and corrugated metal for roofs. 20% of the people have access to electricity. Other major problems include lack of toilet facilities (1 toilet per 50 shacks), lack of water supply, medical facilities, increase in the usage of illegal drugs, unemployment rate of almost 50%, and increased disagreement between tribes.



Manila, PHILIPPINES

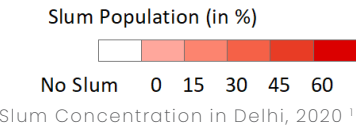
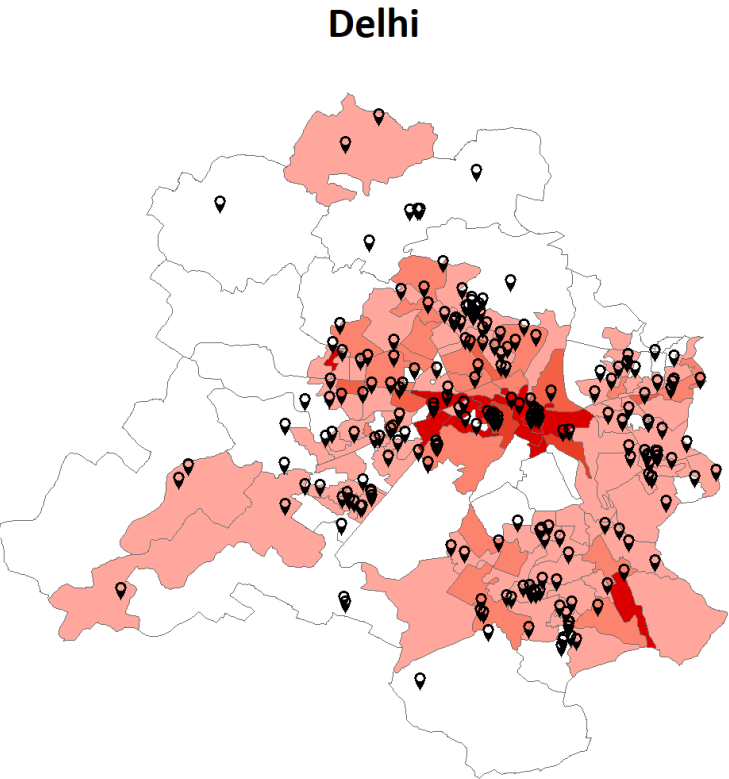
35% (3.4 million) of Manila's population lives in slums. The slums in Manila are home to some of the highest rates of childhood malnutrition in the world (50% of children are found to be anemic). Other major issues include lack of sanitation (66% people do not have access to proper waste disposal, and only 16% children have access to clean water), increase in diseases due to lack of drainage, and vulnerability to natural disasters due to temporary and unstable infrastructure.



Dharavi, Mumbai, INDIA

It is regarded as one of the third largest slums in the world with a population of 1.2 million people. Like other slums in the world, sanitation is one of the key issues that is prevalent here as well. There is an average of 1 toilet for every 1,450 people. Furthermore, while most slums around the globe have high rates of unemployment, the Dharavi slum despite its challenges produces a revenue of 650 USD/yr making it the most productive slum in the world. The goods produced here are sold worldwide, and people mostly run their business on digital platforms.

.slums : a growing concern in the modern day city



THE MILLIONS ON THE MARGINS

▶ In our world, one in eight people live in slums. In total, around a billion people live in slum conditions today

IMPORTANT NUMBERS FROM INDIA

- 44% of slums are situated on private land
- 6% on rail land
- 23% are along drains

MAHARASHTRA HAS...

- 7,723 slums
- 23% of
- 33,510 slums in India

NO SPACE, PRIVACY OR WATER TO DRINK

- ▶ Most slum dwellers still have no security of tenure and live under the constant threat of eviction
- ▶ Access to clean water is also a daily battle for many. In 2006, around 1.1 billion people, globally, had no access to improved water supplies and 2.6 billion people had no access to any type of improved sanitation facility
- ▶ Space is also a problem. In Nairobi, slums cover just 6% of the total residential land area. But they house 60% of the city's population
- ▶ Privacy is a luxury. A 2012 survey shows the average room occupancy in slum households is four, one more person than the recommended maximum, when considering typical slum rooms of nine square meters
- ▶ Permanent constructions are only 25% of all slum structures

DELHI DEMOLITION: 200 ILLEGAL SLUMS DEMOLISHED IN ASHOK VIHAR

NDTV

During the demolition drive in Jahangirpuri on Wednesday. Anand Mohan

DAYS AFTER COMMUNAL CLASH IN DELHI, DEMOLITION BY BJP-RUN CIVIC BODY

Bulldozers roll, raze in Jahangirpuri in the face of Supreme Court order

Court orders status quo but demolition continues for two hours before pause

EXPLAINED Questions over due process

THE ACTION In Jahangirpuri, in the face of the civic body's usual practice of sending a notice at least five days in advance in cases of encroachment. The bylaws state that the notice has to be given to the person who has carried out the alleged illegal construction. During this time, the person can be asked to make alterations, failing which the civic body has the power to demolish or seal the structure.

THE DEMOLITION started around 10.15 am, went on for two hours despite the Supreme Court's direction at 11 am that status quo be maintained. The matter will be heard by a bench of Justices L. Nageswara Rao and B.R. Gavai Thursday.

The demolition action came a day after the civic body wrote to the North West DCP requesting 400 police personnel for what it called was an "anti-encroachment drive". This came after Delhi BJP president Adesh Gupta had also written to the North West DCP Tuesday demanding that "illegal encroachment done by rioters be bulldozed".

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The Indian EXPRESS Thu, 21 April 2022 <https://epaper.indianexpress.com/c/67563497>

WHERE RESIDENTS COMPETE FOR TAPS AND TOILETS

Total number of slums | 3,092

Large or major slums | 23 (where population is 10,000 or more)

Slum population in Kolkata | 18 lakh

Names of some of the major slums | Belgachhia, Cossipore, Dhapa, Dharapara, Tiljala, Pearabagan (Off Sarat Bose Road), Jhoro slum (Tollygunge)

Toilet ratio in a large slum | 1: 25

Roadside water tap ratio in a large slum | 1: 600-700

There are slums where 10-12 people share a 250 square feet room. What can you do in a situation such as this?

Swapan Samaddar | AMIC, SLUM DEVELOPMENT

TOI | APRIL 14, 2020

Congested Belgachhia slum emerges as ground zero of Covid battle in city

An awareness poster near Cossipore slum

File pictures of Nonadanga (left) and Ultadanga slums

.delhi's major slums : a focus on Sangam Vihar

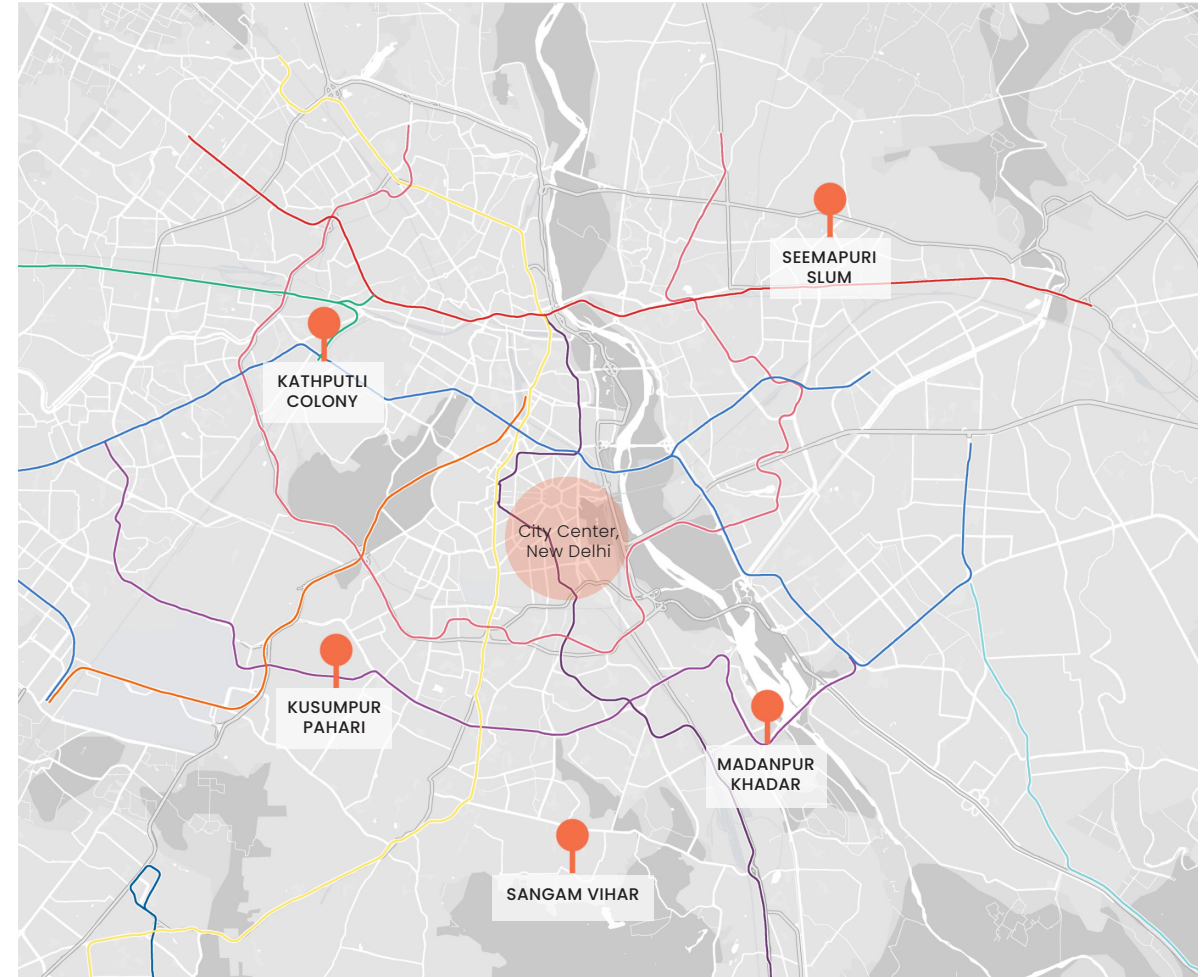
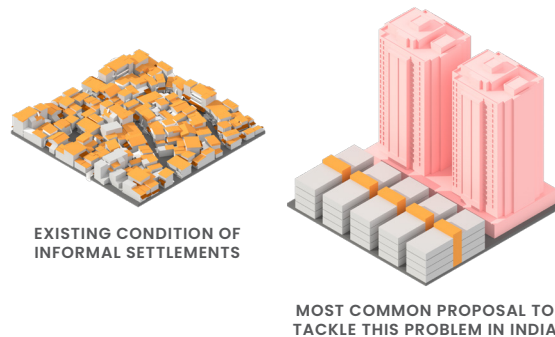
A “dingy backyard” quite perfectly can be used to describe the area of Sangam Vihar due to constant neglect of this unauthorized zone. This area lies in the outer part of Delhi, and is considered to be one of the least prestigious but the fastest growing political constituency.

Despite people having lived there for many years, the risk of demolition has made people build fairly modest one, occasionally two, storey houses in mud or brick. Families of five and six live in one or two rooms houses, more congested than Indian health norms permit. Despite constituting an urban living area for almost three decades, Sangam Vihar has no proper road system and few paved roads, no drainage, sewerage system or garbage collection, and most people rely on private or collective solutions for water supply, electricity, health and education. The area is flooded and muddy during the rains with smell of cloak and uncollected garbage. The municipal government, reflecting unwillingness to invest limited city resources in the development of illegally occupied land, and that demolition was and is its preferred strategy, having neglected the slum

My thesis focuses on addressing the primary infrastructural challenges when it comes to housing in the region of Sangam Vihar.

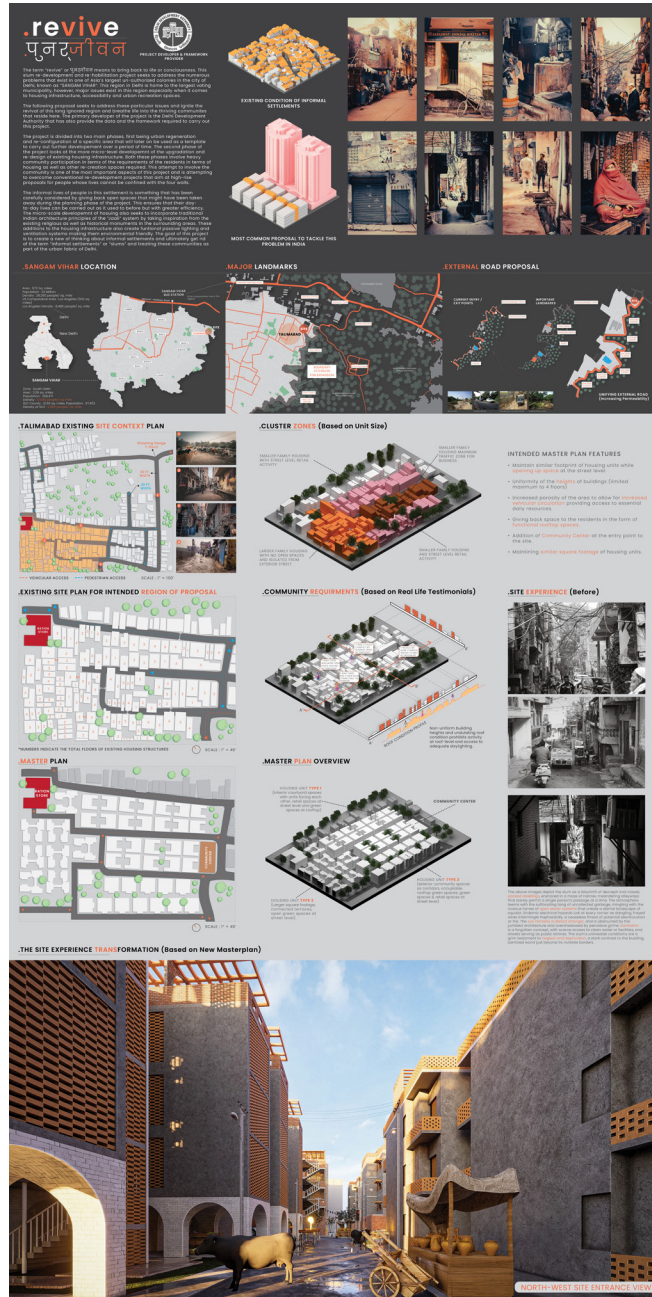
The aim is to re-habilitate and transform this region into a vibrant community that weaves itself together with the existing urban fabric of Delhi.

The approach involves not tearing down this informal settlement but to surgically intervene in order to retain the personality of this region.



Why a Website ...?

.the traditional way of representation : efficient but rigid



.the poster

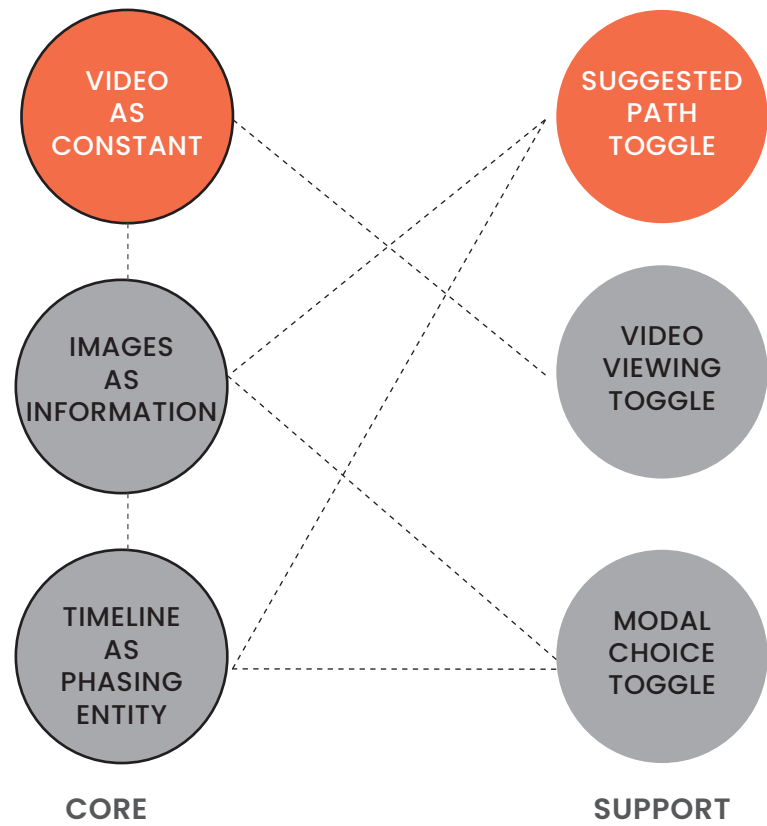
- While the poster is an efficient and creative way of representing an architecture project, it creates an obstacle where it can only incorporate static media.
- A key component of my project relies on the viewers to see the existing state of Sangam Vihar while glancing at the proposals that would benefit this region.
- Additionally, the poster presents all the information at once whereas the project requires the information to be released in phases to the viewer in order to understand the flow of the project as it would appear to be carried out in real life.
- Therefore, the medium of using a website to represent this project would allow me not only to incorporate a diverse set of media but would also allow the viewer to interact with the content in a way that would give a deeper glimpse into the flow of slum re-habilitation projects.

.the concept : parallel narratives

“This website introduces the **Parallel Narratives** approach—a digital documentation method that presents urban rehabilitation through simultaneous formal and informal lenses, allowing stakeholders to navigate between structured planning narratives and organic spatial exploration, thereby democratizing access to complex urban information and proposals.”

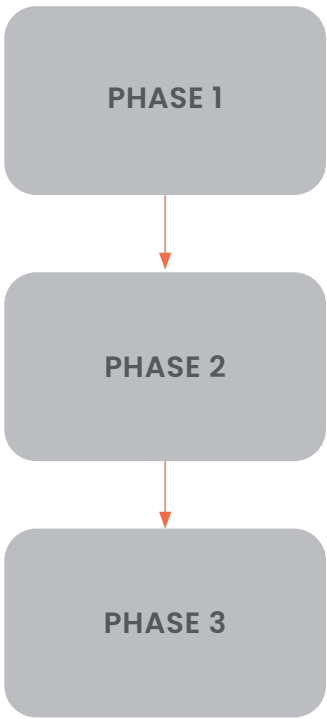
The website incorporates the very ideal of engagement and participation by incorporating ways people can interact with the different type of media. This directly translates to how the users/residents of Sangam Vihar actively participated in the framework of this project.

Media and UI Elements:



Website Structure:

Story Mode (Linear)

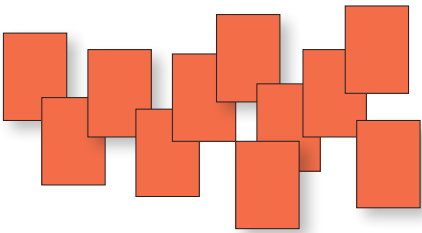


This mode incorporates a top-down approach laying down the purpose and the intended development framework of the project.

This mode acts as the entry point to the visual elements supporting that framework.

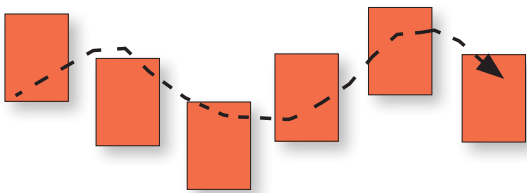
This mode allows the users to familiarize the overall project through text and constantly playing media that can further be investigated in explore mode.

Explore Mode (User-Intended)



FREE-EXPLORE (Stacked Information)

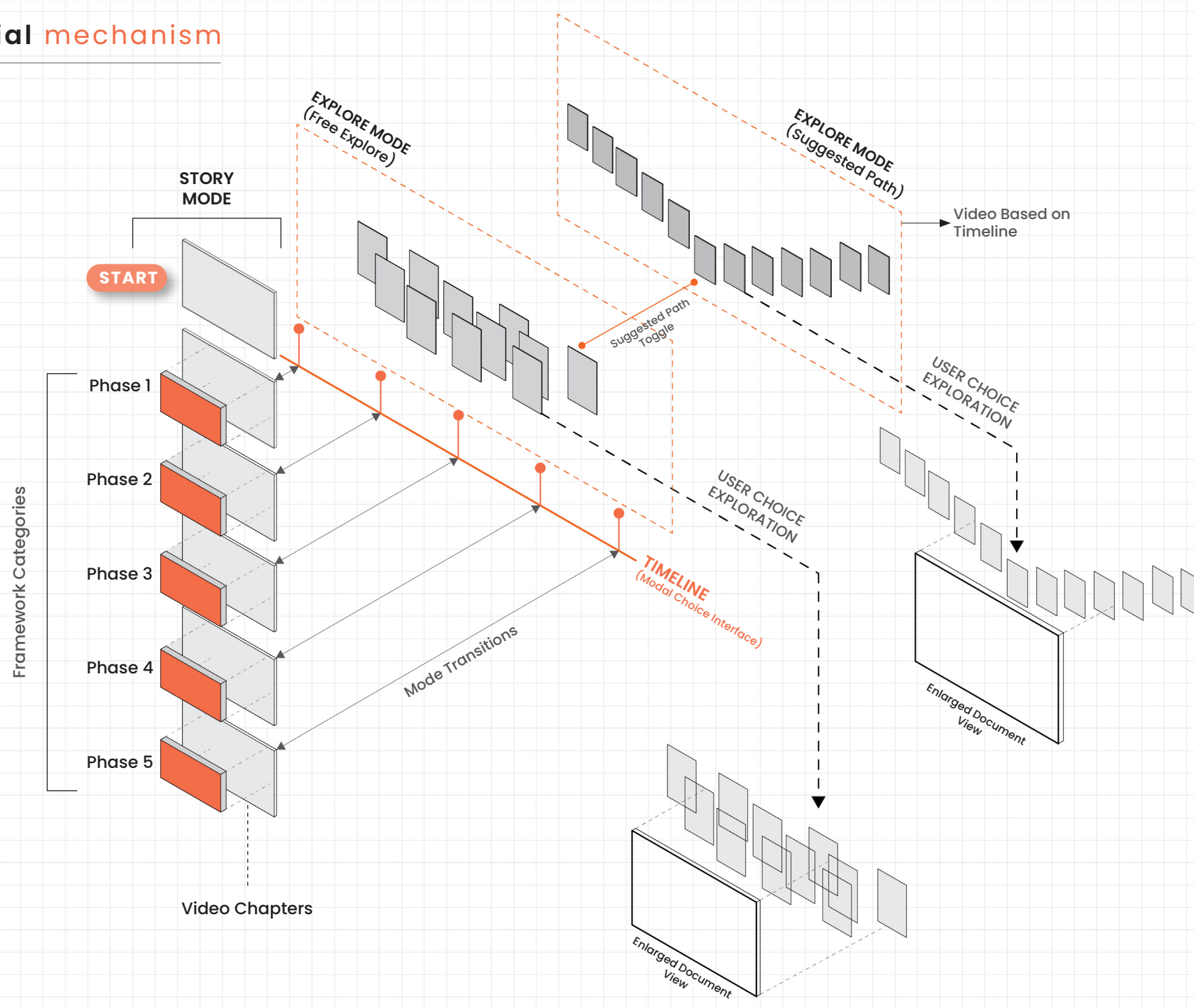
This mode in the ‘explore-mode’ allows users to navigate through images and drawings supporting each phase of the development framework. The stack creates a spatial experience on top of the videos that create immersive environments to convey the sensory reality of Sangam Vihar.



GUIDED-EXPLORE (Chronological Information)

This mode in the ‘explore-mode’ allows users to navigate through the information in a suggested pathway manner as it was intended by developers. This mode is particularly useful for phases that involve a lot of information and need systematic sorting to convey the timeline of the project.

.spatial mechanism



.future potential

Immediate Expansion:

- Comments sections for resident input
- Progress tracking timeline for the project giving residents first-hand information.
- Increased database capacity to store more information for each phase.

Re-usability Framework:

- Template can be adopted for multiple slum re-habilitation projects.
- Video backgrounds can help incorporate site-visits as well as potential proposal experiences.

Long-Term Vision:

- Policy tool template to be used to present to policy-makers.
- Educational use for studying about slum rehabilitation and upgradation.
- Easy-to-use interface allows for easy access on multiple devices, eventually becoming a slum rehabilitation framework app.